Solution

Problems: Difference and Differential Equation Descriptions for Systems

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Difference Equation Descriptions for Systems

1. A system is described by the difference equation

$$y[n] + (9/16)y[n-2] = x[n] + 3x[n-1]$$

Recall that the system function is a ratio of polynomials, $H(z) = b(z)/a(z)$.

a) Find $b(z)$ and $a(z)$.

$b(z) = 1 + 3z^{-1}$

$a(z) = 1 + 9/16\, z^{-2}$

b) Suppose the input to this system is $x[n] = \cos((\pi/2)n)u[n]$. Find the steady-state response of the system.

$x[n] \longrightarrow \cos(\pi/2\, n) = \frac{1}{2} e^{j\pi/2 n} + \frac{1}{2} e^{-j\pi/2 n}$

$y_s[n] = \frac{1}{2} H(e^{j\pi/2}) e^{j\pi/2 n} + \frac{1}{2} H(e^{-j\pi/2}) e^{-j\pi/2 n}$

Note: $e^{j\pi/2} = j$, $e^{-j\pi/2} = -j$

$H(e^{j\pi/2}) = \frac{1-3j}{1-9/16} = \frac{16}{7}(1-3j)$ $\quad H(e^{-j\pi/2}) = \frac{16}{7}(1+3j) = H^*(e^{j\pi/2})$

$y_s[n] = \text{Real}\left\{ \frac{16}{7}(1-3j) e^{j\pi/2 n} \right\}$ $\quad \left(\frac{1}{2} H(e^{j\pi/2}) e^{j\pi/2 n} + \frac{1}{2} \left[H(e^{j\pi/2}) e^{j\pi/2 n} \right]^* \right) = \text{Real}\left\{ H(e^{j\pi/2}) e^{j\pi/2 n} \right\}$

$= \frac{16}{7}\cos(\pi/2 n) + \frac{48}{7}\sin(\pi/2 n)$

2. A system is described by the difference equation

$$y[n] + (1/4)y[n-1] - (1/8)y[n-2] = x[n] + x[n-1]$$

Let the input be $x[n] = (-1)^n u[n]$ and the initial conditions $y[-1] = 4, y[-2] = -2$. Find the output of the system. Identify the transient and steady-state components of the output.

$x[n] \to (-1)^n \Rightarrow y_s[n] = H(-1)(-1)^n$

$H(z) = \dfrac{1+z^{-1}}{1+\frac{1}{4}z^{-1}-\frac{1}{8}z^{-2}}$ $\quad H(-1) = 0$ so $y_s[n] = 0$

Char. Eqn $1+\frac{1}{4}z^{-1}-\frac{1}{8}z^{-2} = 0 = \left(1+\frac{1}{2}z^{-1}\right)\left(1-\frac{1}{4}z^{-1}\right)$

roots: $z_1 = -\frac{1}{2}$, $z_2 = \frac{1}{4}$

$y[n] = 0 + y_t[n] = C_1\left(-\frac{1}{2}\right)^n + C_2\left(\frac{1}{4}\right)^n \quad n \geq 0$

Translate initial conditions to $n \geq 0$

$y[0] = x[0] + x[-1] - \frac{1}{4}y[-1] + \frac{1}{8}y[-2]$

$= 1 + 0 - 1 - \frac{1}{4} = -\frac{1}{4}$

$y[1] = x[1] + x[0] - \frac{1}{4}y[0] + \frac{1}{8}y[-1]$

$= -1 + 1 + \frac{1}{16} + \frac{1}{2} = \frac{9}{16}$

$C_1 + C_2 = -\frac{1}{4}$

$-4\left(-\frac{1}{2}C_1 + \frac{1}{4}C_2 = \frac{9}{16}\right)$

$3C_1 = -\frac{10}{4}$

$C_1 = -\frac{5}{6}$

$C_2 = -\frac{1}{4} + \frac{5}{6} = \frac{7}{12}$

$y[n] = -\frac{5}{6}\left(-\frac{1}{2}\right)^n + \frac{7}{12}\left(\frac{1}{4}\right)^n, \quad n \geq 0$

Differential Equation Descriptions for Systems

3. A system is described by the differential equation

$$\frac{d^2}{dt^2}y(t) + 6\frac{d}{dt}y(t) + 8y(t) = 2x(t)$$

a) Find the transfer function of the system, $H(s) = b(s)/a(s)$.

$$H(s) = \frac{2}{s^2+6s+8}$$

b) Suppose the input to the system is $x(t) = e^{-t}u(t)$ and the initial conditions are $y(t)|_{t=0^-} = -1$, $\frac{d}{dt}y(t)\big|_{t=0^-} = 1$. Find the output of the system. Identify the transient and steady-state components.

$x(t) \to e^{-t} \Rightarrow y_s(t) = H(-1)e^{-t}$, $H(-1) = \frac{2}{3}$

Char Eqn: $s^2+6s+8 = 0 = (s+2)(s+4)$

roots: $s_1 = -2$, $s_2 = -4$

$y_t(t) = c_1 e^{-2t} + c_2 e^{-4t}$

$y(t) = \frac{2}{3}e^{-t} + c_1 e^{-2t} + c_2 e^{-4t} \quad t > 0$

Init. cond. at $t = 0^-$ also apply at $t = 0^+$ (no impulses on RHS for this $x(t)$)

$y(0) = \frac{2}{3} + c_1 + c_2 = -1$ ⇒ $c_1 + c_2 = -5/3$ × 2

$\frac{d}{dt}y(t)\big|_{t=0^+} = 1 = -2/3 - 2c_1 - 4c_2$ ⇒ $-2c_1 - 4c_2 = 5/3$

$-2c_2 = -5/3$, $c_2 = 5/6$

$c_1 = -5/3 - 5/6 = -\frac{15}{6} = -\frac{5}{2}$

$y(t) = \underbrace{\frac{2}{3}e^{-t}}_{\text{steady state}} \underbrace{- \frac{5}{2}e^{-2t} + \frac{5}{6}e^{-4t}}_{\text{transient}}, \; t > 0$

MATLAB Exploration

4. In this problem we use MATLAB to compute the output of two systems described by difference equations.

System 1: $y[n] - 2.7429y[n-1] + 2.5411y[n-2] - 0.7933y[n-3] = 0.0147x[n] - 0.0123x[n-1] - 0.0123x[n-2] + 0.0147x[n-3]$

System 2: $y[n] - 2.1249y[n-1] + 1.6087y[n-2] - 0.3954y[n-3] = 0.6489x[n] - 1.9156x[n-1] + 1.9156x[n-2] - 0.6489x[n-3]$

The MATLAB code DiffEqnProbs will play a clip of music and the clip after being processed by both System 1 and System 2. The script also finds the roots of the characteristic equations for each system (displayed in the command window), plots the impulse response of each system, and plots the response of each system to unit amplitude low- and a high-frequency cosine input signals.

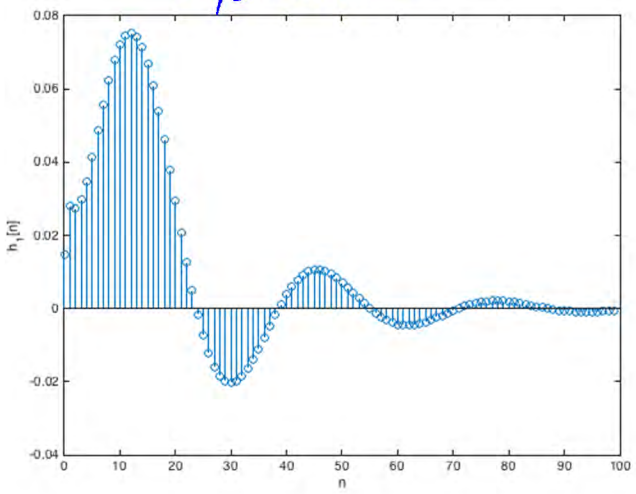
a) System 1 tends to attenuate (low, high) frequencies while System 2 tends to attenuate (low, high) frequencies.

b) Based on the roots of the characteristic equation, which system is likely to have the fastest transient response?

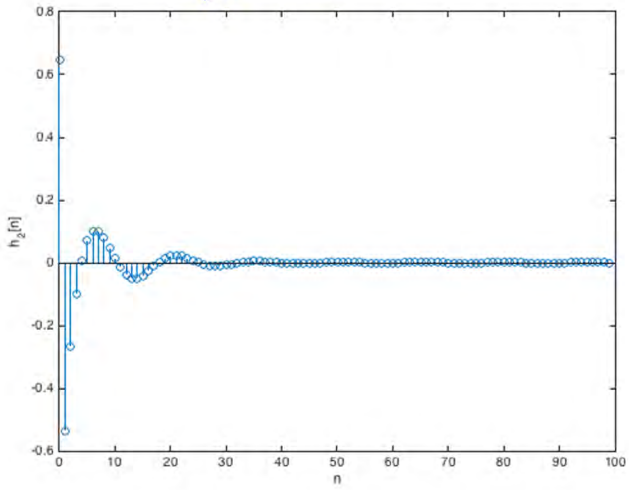
Root d_i contributes d_i^n to transient response.
This term decays as $|d_i|^n$.
All roots of System 2 are smaller in magnitude than the corresponding roots of System 1, so System 2 should have faster transient response.

c) Which system has the shortest duration response to an impulse input? Does this agree with your conclusions based on the magnitude of the roots of the characteristic equation?

System 1



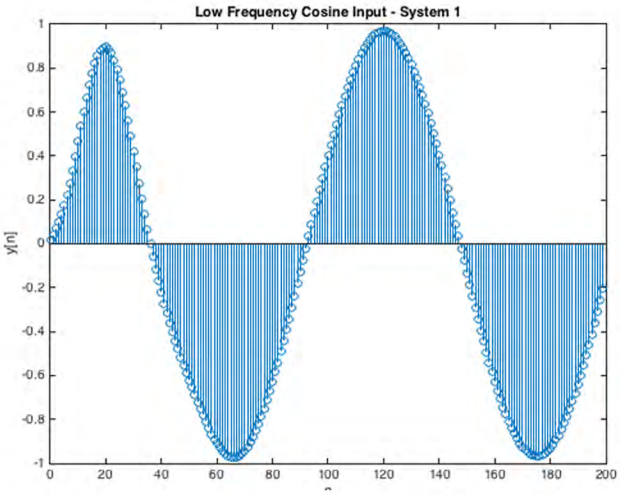
System 2

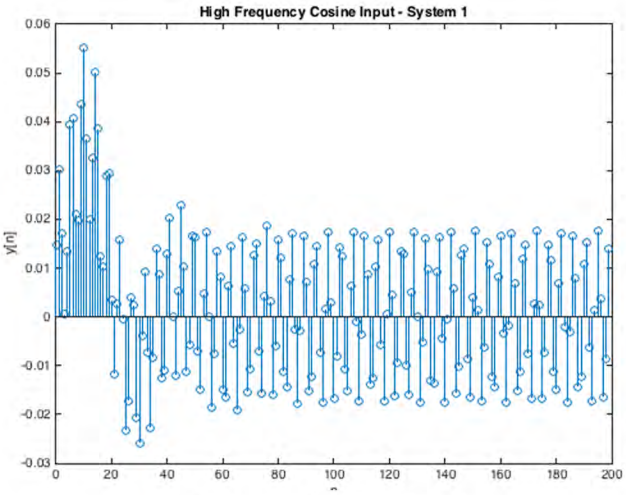


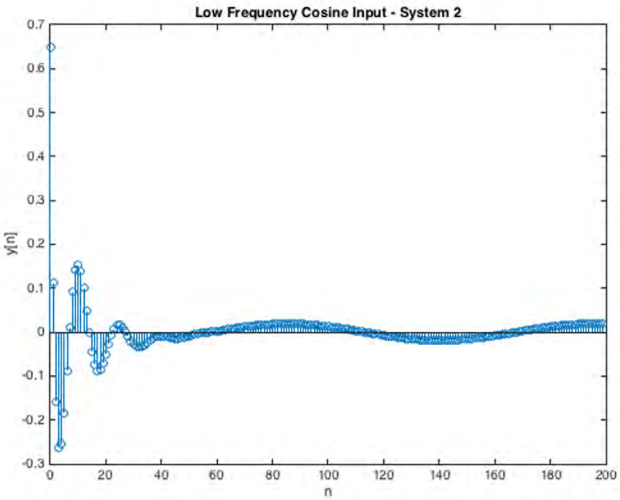
The impulse response of System 2 appears to have the shorter duration, comparing h[n] to its max value.

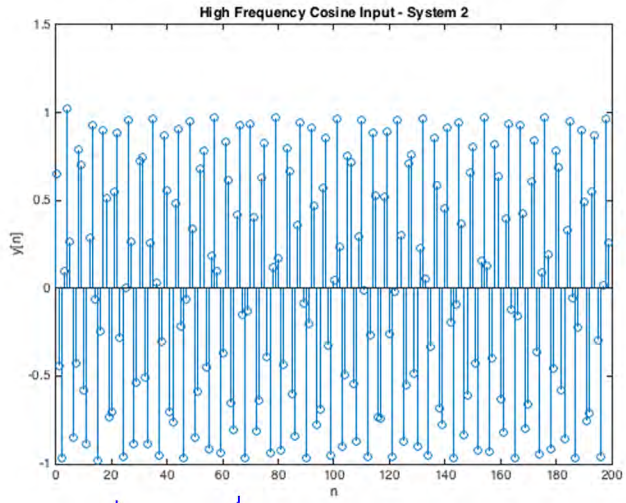
Yes this agrees with the prediction based on char. eqn. roots

d) Evaluate the steady-state gain of the each system to the low- and high-frequency cosines.









input sinusoid has unit amplitude, so gain is given by output sinusoid amplitude

i) Gain of System 1 to the low-frequency cosine: ~1

ii) Gain of System 1 to the high-frequency cosine: ~0.02

iii) Gain of System 2 to the low-frequency cosine: ~0.02

iv) Gain of System 2 to the high-frequency cosine: ~1

e) Are the gains of each system to the low- and high-frequency cosine in agreement with what you hear when listening to the music clip?

Yes!